

AI Job Copilot V2.1 Release Report

Objective

Move the slowest long-running portfolio and job-application workflows to additive background-job endpoints

Baseline timing

- Synchronous portfolio baseline (portfolio_devops): 160120.09 ms
- Synchronous job application baseline (job_application_devops): 160457.99 ms

New architecture

- Added in-memory background job manager, store, runner, lifecycle models, and expiry handling.
- Added additive background endpoints for portfolio and job-application creation, polling, result retrieval, a
- Kept synchronous endpoints intact by routing both sync and async paths through shared workflow helpers.
- Added background-job analytics endpoint protected by the existing admin secret.

Performance comparison

- Background portfolio create latency: 12.51 ms
- Background portfolio total processing: 114759.76 ms
- Background job application create latency: 6.47 ms
- Background job application total processing: 181586.65 ms

Files changed

- C:\Users\surya\AI-Jobs\main.py
- C:\Users\surya\AI-Jobs\background_jobs__init__.py
- C:\Users\surya\AI-Jobs\background_jobs\job_models.py
- C:\Users\surya\AI-Jobs\background_jobs\job_errors.py
- C:\Users\surya\AI-Jobs\background_jobs\job_store.py
- C:\Users\surya\AI-Jobs\background_jobs\job_runner.py
- C:\Users\surya\AI-Jobs\background_jobs\job_cleanup.py
- C:\Users\surya\AI-Jobs\background_jobs\job_manager.py
- New backend tests under C:\Users\surya\AI-Jobs\tests\test_background_*.py and job lifecycle tests.

Job lifecycle behavior

- Jobs now move through queued, processing, completed, failed, cancelled, and expired states.
- Progress stages are monotonic and exposed through GET /jobs/{job_id}.
- Results are retrieved through GET /jobs/{job_id}/result with safe 202, 410, and failure behavior.

Backward compatibility

- Existing synchronous endpoints /generate-portfolio and /prepare-job-application remain present in OpenAPI
- Automated tests confirmed synchronous route compatibility.

Testing

- Automated backend tests passed: 17
- Real synthetic workflow checks completed: 6

Known issues

- Background jobs are currently in-memory only; they are designed for future queue backends but do not yet
- Admin background-job analytics are operational summaries of the current in-memory store, not long-term p

Artifacts

- Baseline folder: C:\Users\surya\AI-Jobs\performance_regression\baseline_v2_1
- After folder: C:\Users\surya\AI-Jobs\performance_regression\after_v2_1
- Manual review folder: C:\Users\surya\AI-Jobs\performance_regression\after_v2_1\manual_review

Acceptance result

V2.1 status: PASS WITH ISSUES

Reason: core background workflow reliability is complete and validated, with only future infrastructure persis